

SODANKYLA, Finland

WMO: 028360

Lat: 67.367N

Lon: 26.629E

Elev: 602

StdP: 14.38

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.8	-23.4	-33.6	0.8	-28.2	-28.1	1.2	-23.1	18.6	23.6	16.9	23.6	2.6	310	0.693

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.9	77.3	61.4	73.3	59.6	69.7	58.1	64.3	72.9	61.9	69.7	59.8	66.9	6.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.1	82.1	67.3	58.8	75.6	64.9	56.7	70.0	62.7	29.7	72.5	28.0	69.7	26.6	67.0	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.2	13.4	11.8	DB	-35.0	81.9	6.2	3.7	-39.5	84.6	-43.1	86.7	-46.6	88.8	-51.2	91.5	
			WB	-34.6	67.0	5.7	2.7	-38.8	69.0	-42.1	70.6	-45.3	72.1	-49.5	74.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	32.1	8.3	8.9	18.2	29.8	41.5	52.6	58.7	54.1	44.7	32.5	20.5	13.4
	DBStd	20.32	15.16	14.97	11.26	8.57	8.00	7.12	6.30	6.41	6.72	8.50	11.93	14.71
	HDD50	7153	1292	1150	985	605	287	52	6	31	183	543	885	1134
	HDD65	12044	1757	1570	1450	1055	729	377	213	343	608	1008	1335	1599
	CDD50	609	0	0	0	0	22	129	275	157	26	0	0	0
	CDD65	27	0	0	0	0	0	5	17	5	0	0	0	0
	CDH74	282	0	0	0	0	11	57	168	46	0	0	0	0
	CDH80	42	0	0	0	0	0	8	28	6	0	0	0	0
Wind	WSAvg	5.5	5.0	5.3	6.1	5.6	6.1	5.9	5.5	5.3	5.5	5.3	5.1	5.3
Precipitation	PrecAvg	21.0	1.4	1.2	1.1	1.1	1.5	2.3	2.8	2.4	2.1	1.9	1.7	1.5
	PrecMax	30.9	2.8	2.8	2.6	3.1	4.0	4.4	5.5	5.4	4.6	4.6	2.8	3.0
	PrecMin	16.1	0.2	0.2	0.2	0.3	0.2	0.2	0.3	0.7	0.2	0.2	0.3	0.4
	PrecStd	3.3	0.6	0.6	0.6	0.6	0.7	1.1	1.2	1.1	1.0	0.9	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.4	37.6	43.0	56.3	75.2	80.9	83.5	79.3	65.2	51.5	42.6	38.1
		MCWB	33.9	34.8	36.3	45.4	56.5	62.0	64.2	64.2	55.7	48.6	41.6	35.5
	2%	DB	32.8	33.3	38.5	49.2	67.3	73.8	78.7	73.8	61.3	48.8	38.5	35.1
		MCWB	31.8	31.8	33.7	40.5	53.5	58.0	62.8	60.6	53.8	46.5	37.3	34.0
	5%	DB	30.4	31.6	35.1	45.2	60.5	69.6	74.7	69.5	58.3	46.4	35.7	33.1
		MCWB	29.5	30.6	31.7	38.2	49.2	56.3	60.8	58.8	52.7	44.2	35.1	32.2
	10%	DB	27.7	29.1	32.8	41.9	55.7	65.7	71.0	65.7	55.3	43.3	33.8	31.3
		MCWB	27.0	28.0	30.5	36.0	46.7	54.1	59.5	57.4	51.6	41.7	33.1	30.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.1	35.2	37.0	45.9	57.1	64.4	67.5	66.3	57.8	49.8	42.0	36.0
		MCDB	35.4	37.2	42.4	55.0	70.8	75.0	75.4	75.7	62.1	50.7	42.3	37.1
	2%	WB	32.0	32.3	34.2	41.1	54.2	60.8	65.2	63.1	55.4	47.0	37.6	34.1
		MCDB	32.7	33.2	37.4	47.3	65.8	69.7	74.4	70.2	59.4	48.4	38.4	34.8
	5%	WB	29.6	30.5	32.5	38.8	50.6	57.8	63.1	60.3	53.7	44.4	35.2	32.4
		MCDB	30.2	31.6	34.6	44.1	58.9	66.2	71.1	66.6	57.1	46.1	35.7	33.1
	10%	WB	27.0	28.0	30.8	36.7	47.5	55.5	61.1	58.3	51.9	41.9	33.3	30.5
		MCDB	27.7	29.0	32.8	40.9	54.5	63.5	68.5	64.0	54.8	43.3	33.7	31.2
Mean Daily Temperature Range		MDBR	15.2	16.0	19.1	17.2	15.8	15.5	15.9	16.1	13.2	9.1	10.8	13.9
	5% DB	MCDBR	11.8	11.1	16.7	19.6	25.7	22.8	23.1	22.7	18.2	9.6	6.8	8.7
		MCWBR	11.7	10.3	13.2	12.3	13.6	9.8	9.9	11.1	11.5	7.9	6.8	8.4
	5% WB	MCDBR	12.2	10.9	13.6	17.4	22.9	18.7	18.9	18.6	14.7	8.8	6.7	8.5
		MCWBR	12.1	10.3	11.2	11.3	13.0	9.5	9.8	10.2	9.9	7.8	6.7	8.4
Clear-Sky Solar Irradiance	taub	0.193	0.257	0.291	0.323	0.340	0.341	0.354	0.336	0.310	0.277	0.351	N/A	
	taud	1.924	2.093	2.162	2.206	2.422	2.474	2.454	2.483	2.536	2.328	2.710	N/A	
	Ebn at Noon	47	176	237	262	272	274	266	259	239	174	57	0	
	Edh at Noon	4	16	26	33	30	30	29	25	19	13	4	0	
All-Sky Solar Radiation	RadAvg	16	173	652	1247	1528	1612	1476	1079	559	187	35	0	
	RadStd	1	24	53	86	136	133	150	100	56	36	5	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.26	N/A	N/A	+1.48	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (37 neighbors)	+1.12	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page